1 Establish matched comparisons

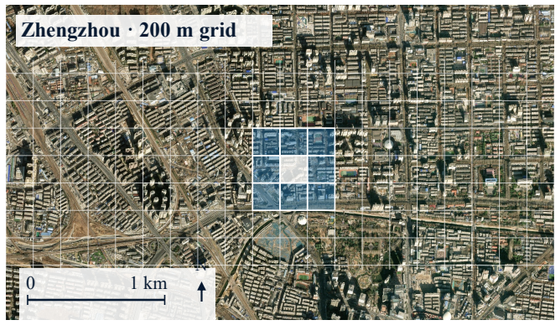
Beyond own attributes and ordinary adjacency?

Own attributes (no graph)



Each location uses only its own attributes

Ordinary adjacency (undirected)



Information from local neighbouring locations

Domain-informed graph (D8, terrain flow)



D8

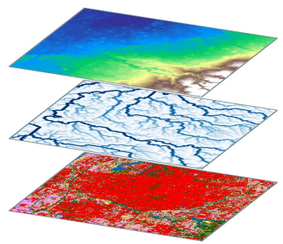
200 m grid

Upstream-to-downstream information from terrain flow

2 Examine information conditions

Which information and predictor conditions matter?

Attribute availability



Topographic attributes

Hydrological attributes

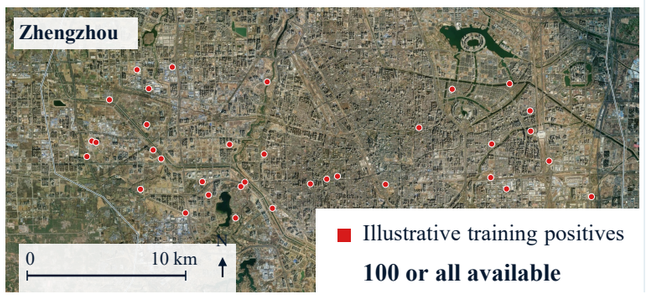
Land use attributes

Socio-environmental attributes

18 / 16 / 14 attributes

Attribute × budget series: Zhengzhou, two-layer GraphSAGE

Training-positive budget

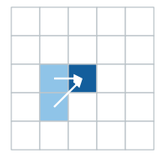


Illustrative training positives

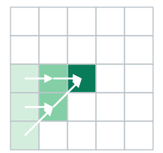
100 or all available

Predictor and receptive field

Fixed propagation
(one step + FFN)



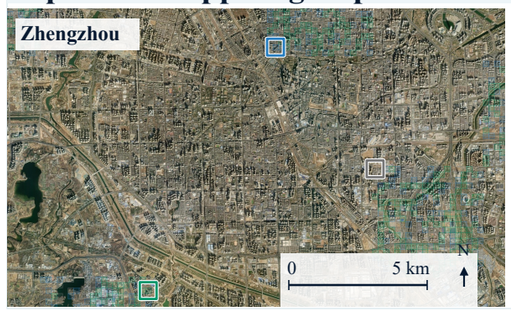
GraphSAGE
(one or two layers)



3 Locate gains and losses

Where do graph gains and losses arise?

Upstream-support groups on a terrain surface



No upstream

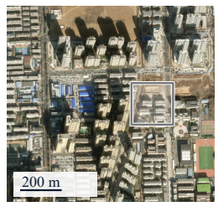
One hop only

Two or more hops

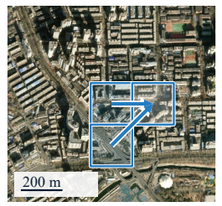
Held-out graph (test fold 3)

Group contributions to the overall AP change

No upstream



One hop only



Two or more hops



Group contributions sum to the total AP change

Background-relative rank change

Recorded positives relative to the background

Lower rank — Higher rank



Recorded flood cell

4 Assess stability

Do judgments persist across evaluation choices?

Evaluation metrics

Sampled AP
(three fixed 1:10 samples)



Sample background

Full-domain AP
(all eligible test cells)



Retain all background

Reweighted AP
(same cells and scores)



Change background weight

Recorded positive

Unlabelled background

Same fitted predictions

Alternative evaluation settings

Background weight

w = 1

w = 10M/|B|

Unit weight or reweighted unlabelled background

Inspection capacity



Top-scoring 10% per test fold

Recall of recorded positives